Statistical Arbitrage Strategy

- **Introduction:**
  Statistical Arbitrage (Stat Arb) is a quantitative trading strategy that uses mathematics, statistics, and historical market relationships to identify temporary price inefficiencies in the market.
  The main idea behind Statistical Arbitrage is:
  If related stocks move abnormally far apart, they may return back to normal later.
  Traders try to profit from this temporary imbalance.
  Statistical Arbitrage is widely used by:
    - hedge funds
    - quantitative traders
    - algorithmic trading firms

  because it focuses on mathematical probability instead of emotional trading.

- **Core Concept of Statistical Arbitrage:**
  Some stocks or assets usually move together because:
    - they belong to the same sector
    - have similar business models
    - react similarly to market conditions

  If their price relationship becomes abnormal temporarily, traders expect the relationship to normalize again.

- **Important Statistical Concepts:**

Concept	**Meaning**
Spread	Difference between two prices
Mean	Average value
Correlation	Stocks moving similarly
Cointegration	Long-term stable relationship
Z-Score	Distance from average

- **Types of Statistical Arbitrage Strategies:**

1. Pair Trading Strategy:

- Pair Trading is the most common Statistical Arbitrage strategy.
- It involves two correlated stocks that usually move together.
- If the price gap between them becomes unusually large, traders expect the gap to reduce later.

Formula — Spread:

$$Spread = Price_A - Price_B$$

Explanation of Terms:

Term	Meaning
Price_A	Price of first stock
Price_B	Price of second stock
Spread	Difference between prices

Example:

Suppose:

Stock	Price
Visa	140
Mastercard	110

Step 1 — Calculate Spread

$$Spread = 140 - 110 = 30$$

Suppose normal spread is usually:
10
Current spread:
30
This means the stocks moved abnormally far apart.

Trading Decision:

Trader:

- SELLS Visa
- BUYS Mastercard

expecting prices to normalize.

Best For:

- Correlated stocks
- Same industry companies

Advantages:

- Market-neutral strategy
- Lower market risk
- Widely used in quant trading

Disadvantages:

- Correlation may break

- Requires statistical analysis

2. Mean Reversion Arbitrage:
This strategy assumes that if price moves too far from its average value, it may eventually return back toward the average.

Formula — Moving Average:

$$MA = \frac{P_1 + P_2 + \ldots + P_n}{n}$$

Explanation of Terms:

Symbol	Meaning
P_1, P_2	Previous prices
n	Number of prices
MA	Average price

Example:

Suppose last 5 prices are:

Prices
100
102
98
101
99

Step 1 — Calculate Average Price:

$$MA = \frac{100 + 102 + 98 + 101 + 99}{5} = 100$$

Current stock price becomes: 130
Price moved too far above average.
Trader expects: price may return toward 100
Possible: **SELL**

Best For:

- Sideways markets
- Stable markets

Advantages:

- Simple to understand
- Popular in quantitative trading

Disadvantages:

- Strong trends can continue longer
- False reversals possible

3. Z-Score Mean Reversion Strategy:

Z-Score measures how far the current price is from its average value using standard deviation.

Formula — Z-Score:

$$Z = \frac{X - \mu}{\sigma}$$

$$x$$

$$\mu$$

$$\sigma$$

$$z = \frac{x - \mu}{\sigma} \approx 1.2$$

$$\Phi(z) \approx 88.5\%$$

Explanation of Terms:

Symbol	Meaning
X	Current price
μ	Average price
σ	Standard deviation

Example:

Suppose:

- Current price = 130
- Average price = 100
- Standard deviation = 10

Step 1 — Calculate Z-Score:

$$Z = \frac{130 - 100}{10} = 3$$

Interpretation:

Z-score = 3 means: price is extremely far above normal

Possible: **SELL signal**

Common Rule:

Z-Score	Meaning
Above +2	Overbought
Below -2	Oversold

Best For:

- Statistical trading
- Quantitative systems

Advantages:

- Strong mathematical approach
- Detects abnormal price movement

Disadvantages:

- Requires statistical understanding
- Sensitive to volatility

4. Cointegration Strategy:

Cointegration checks whether two stocks have a stable long-term relationship even if they move differently in the short term.

Formula:

$$Y_t = \alpha + \beta X_t + \epsilon_t$$

Explanation of Terms:

Symbol	Meaning
Y_t	Price of stock Y
X_t	Price of stock X
α	Constant
β	Relationship strength
ε_t	Error/spread

Example:

Suppose:

- Coca-Cola
- Pepsi

usually move together.

Suddenly:

Coca-Cola	Pepsi
150	112

Relationship becomes abnormal.
Trader:

- SELLS Coca-Cola
- BUYS Pepsi

expecting prices to normalize later.

Best For:

- Advanced quant trading
- Long-term statistical relationships

Advantages:

- More reliable than simple correlation
- Used by professional quant firms

Disadvantages:

- Complex statistical calculations
- Relationships may fail

5. Index Arbitrage Strategy:

Index Arbitrage exploits temporary price differences between an index and the actual stocks inside the index.

Formula:

$$Index\ Value = \sum(Stock\ Price \times Weight)$$

Explanation of Terms:

Term	Meaning
Stock Price	Price of individual stock
Weight	Percentage contribution in index

Example:

Suppose index contains:

Stock	Price	Weight
A	100	50%
B	200	50%

Step 1 — Calculate Index Value:

$$Index = (100 \times 0.5) + (200 \times 0.5) = 150$$

Suppose market shows:
Index = 170
Index became overpriced.

Trader:

- SELLS index
- BUYS underlying stocks

expecting normalization.

Best For:

- Institutional trading
- High-frequency trading

Advantages:

- Lower directional market risk
- Widely used by institutions

Disadvantages:

- Requires fast execution
- High transaction cost

6. ETF Arbitrage Strategy:

ETF Arbitrage exploits temporary mismatch between ETF price and the actual value of assets inside the ETF.

Formula — NAV:

$$NAV = \frac{Total\ Asset\ Value - Liabilities}{Number\ of\ Shares}$$

Explanation of Terms:

Term	Meaning
NAV	Net Asset Value
Asset Value	Total value of ETF assets
Liabilities	ETF expenses/debt
Shares	Total ETF shares

Example:

Suppose:

- ETF market price = 105
- Actual NAV = 100

ETF is overpriced.

Trader:

- SELLS ETF
- BUYS underlying assets

expecting prices to normalize.

Best For:

- ETF markets
- Institutional trading

Advantages:

- Efficient pricing opportunities
- Lower directional risk

Disadvantages:

- Requires large capital
- Fast execution needed

➢ **Advantages of Statistical Arbitrage:**

- Uses mathematical probability
- Lower emotional trading
- Market-neutral opportunities
- Widely used in quantitative finance

➢ **Disadvantages of Statistical Arbitrage:**

- Complex statistical analysis
- Relationships may break
- Requires high-quality data
- Transaction costs can impact profits